Fault of Our Stars: Behavioral Drivers of Rating–Sentiment Incongruence

Abstract—**When people share experiences online, they often express thoughts in two ways: a star rating and written review. In sentiment analysis, ratings are widely used as reliable labels for sentiment in that text, yet whether the two actually agree is rarely questioned. This study investigates what happens when they don't, a phenomenon defined as sentiment–rating incongruence, in the context of Sri Lankan tourism attraction reviews and treats such mismatches as meaningful signals of reviewer behavior, not labeling errors. A large dataset of attraction reviews is analyzed using a transformer-based sentiment pipeline that derives textual sentiment independently of assigned rating. Incongruence is found in 18.6% of reviews and falls into six directional patterns, with Conservative Rater and Obligatory 5-Star behaviors making up the majority mismatches. Incongruence also vary noticeably between venue types, with museums showing highest prevalence. A multi-stage analytical framework combining Statistical Tests, Logit Modeling, Random Forest, and SHAP Interpretation is then applied to identify the behavioral factors behind these patterns, pointing to venue type, reviewer expertise, and review length as the strongest drivers. In conclusion, sentiment–rating incongruence is systematic and context-dependent, questioning the reliability of ratings as ground truth and underscoring the need for more context aware sentiment approaches.**

Index Terms—**Sentiment Analysis, Sentiment–Rating Incongruence, Tourism, Behavioral Analysis, Review Analytics**

I. Introduction

Tourism reviews are an important source of information for travel decision-making, and most review platforms present both numerical star ratings and written review text. In sentiment analysis, these ratings are often treated as reference labels because they provide a convenient mapping to sentiment polarity [1], [2]. However, this assumption remains insufficiently examined. In many cases, numerical ratings and textual sentiment do not align, which may introduce systematic noise into downstream models. The rapid growth of user-generated content on platforms such as TripAdvisor has greatly increased the volume of available review data [3], but the reliability of ratings as proxies for textual sentiment has not been examined at a comparable pace.

These variations should not be treated simply as errors. Instead, they may indicate consistent reviewer behavior influenced by expectations, experience, and context. For example, a visitor may describe a museum positively but assign only three stars because the experience did not meet their prior expectations. Conversely, a reviewer might give five stars to a religious site while expressing mixed sentiment in the text because of cultural or social considerations. In this study, this phenomenon is defined as sentiment–rating incongruence and is treated as a structured behavioral pattern rather than a simple labeling problem. Previous studies also suggests that a considerable percentage of reviews show some form of inconsistency between sentiment polarity and rating scores [4], [5], indicating that the assumption of rating–text alignment is lacking a strong validation. Transformer-based methods are particularly well matching to identifying such minor differences when compared with traditional lexicon-based approaches [6], [7].

Despite its importance, sentiment–rating incongruence remains under-studied. Previous research has largely relied on ratings as proxy labels and has focused mainly on hotels and restaurants in well-studied settings [2], [8]. Topic modeling and aspect-based sentiment analysis have improved understanding of how specific service aspects relate to overall ratings [9], [10], yet the broader issue of systematic incongruence across review types and attraction categories has received less attention. In the Sri Lankan context, prior studies have identified rating–text incompatibility in hotel reviews [11], [12], while broader international research points to contextual variation and persistent inconsistency [4], [5], [13].

There is still limited understanding of how common sentiment–rating incongruence is, what forms it takes, and which factors drive it, particularly in tourism attraction reviews from emerging destinations. Reviews of the hospitality literature suggest that attractions have received less attention than hotels and restaurants, especially in underrepresented settings such as Sri Lanka [8]. At the same time, advances in transformer-based and transfer-learning methods make it possible to study this phenomenon at scale without relying on ratings as the sole source of sentiment labels. Using a large dataset of Sri Lankan tourism attraction reviews collected between 2010 and 2023, this study investigates how textual sentiment and numerical ratings diverge and examines the structural and behavioral factors associated with that divergence.

To address these gaps, this study examines sentiment–rating incongruence in a large-scale dataset of Sri Lankan tourism attraction reviews. A transformer-based sentiment pipeline is used to independently derive textual sentiment, enabling the identification of mismatches between sentiment and ratings. These mismatches are organized into directional patterns, and a multi-stage analytical framework combining statistical and machine learning techniques is applied to examine their structural and behavioral drivers.

This study makes four contributions. First, it provides a large-scale empirical analysis of sentiment–rating incongruence in tourism attraction reviews. Second, it introduces a structured typology of mismatch patterns. Third, it identifies key structural and behavioral drivers using an integrated analytical framework.

Finally, it shows that rating–text inconsistency is a meaningful, context-dependent signal with implications for sentiment modeling.

II. Related Work

The foundational survey by Pang and Lee [1] established sentiment analysis as a major research area and reinforced the common assumption that star ratings broadly reflect the sentiment expressed in review text. Despite recognized limitations, this convention remains widely used in tourism research as a practical weak-labeling strategy. Alaei et al. [2] note that ratings are often treated as weak labels without explicit validation, and that the literature has been heavily concentrated on hotels and restaurants. This imbalance is further highlighted by Ameur et al. [8], who report limited venue diversity and restricted geographic coverage in existing studies, especially for emerging tourism destinations.

Methodological advances have substantially improved sentiment analysis in tourism. Wen et al. [6] demonstrate the effectiveness of transformer-based models such as BERT [14] and ERNIE [15], while Puh and Bagic Babac [7] show that jointly analyzing sentiment and ratings can provide more detailed insight. Multilingual approaches and aspect-based methods further improve interpretability by linking sentiment to specific components of the tourism experience [9], [10]. More recently, zero-shot approaches have expanded the feasibility of analyzing under-studied datasets with limited labeled data [16].

This inconsistency shows up in regional literature as well. Abeysinghe and Walgampaya [11] document rating–text incompatibility in hotel reviews in Anuradhapura, while Abeysinghe and Bandara [12] extend this finding across five Sri Lankan cities and propose a self-learning approach to resolve it. However, both depends on lexicon-based methods and define the problem primarily as one requiring correction rather than explanation. However, the present study focuses on tourism attraction reviews, uses transformer-based sentiment analysis, and interprets incongruence as a behavioral phenomenon.

Empirical findings on the relationship between ratings and review text remain mixed. Bigne et al. [4] report general alignment between the two, but also identify variation across contexts. George and Ramos [3] show that ratings may exceed text-based sentiment in destination-related reviews, while Kwon et al. [5] demonstrate that rating–text inconsistency varies by context and influences perceived review usefulness. These findings suggest that ratings and text do not always capture the same dimension of experience.

Reviewer characteristics also appear to matter. Chua and Banerjee [13] show that reviewer expertise influences the relationship between ratings and textual content, while related work links sentiment polarity and review depth to perceived usefulness [8], [13]. Taken together, these studies indicate that ratings and text may encode different aspects of user experience and that inconsistency may be partly shaped by reviewer-level behavior.

Overall, sentiment–rating incongruence remains insufficiently understood, particularly in tourism attraction contexts and emerging destinations. Although recent methods enable large-scale and fine-grained analysis [6], [7], [9], [10], [16], there is still limited evidence on the structure of directional mismatch patterns and their drivers in multi-venue, longitudinal datasets. This study addresses that gap using a large-scale Sri Lankan tourism review dataset spanning diverse attraction types and extended time span [17].

Building on these limitations, this study addresses the gap by providing a large scale, context-specific analysis of sentiment–rating incongruence in tourism attraction reviews from an underrepresented setting. Unlike prior work that treats rating–text mismatch primarily as a labeling issue, this study adopts a behavioral perspective, examining the structure of directional incongruence patterns and their underlying drivers. By combining transformer-based sentiment inference with statistical and machine learning methods, this study explains how contextual and reviewer-level factors shape sentiment–rating divergence.

III. Methodology

TABLE I
Methodology Overview

Ph.	Stage	Method / Tool	Output
1	Dataset and Preprocessing	Rule-based normalization, Feature engineering	Clean review dataset
2	Sentiment Model Selection	Comparative evaluation of RoBERTa variants on 1,000 labeled reviews for optimal model selection	Best performing model selected for full dataset sentiment prediction
3	Variable Construction	Derived from model output and reviewer data	Incongruence flag, Pattern type, Reviewer tier
4	Statistical Testing and Predictive Modeling	Chi-square, Mann–Whitney U, Logit, Random Forest, SHAP	Key drivers and effect sizes

A. Dataset and Preprocessing

The framework begins by transforming a large, venue-diverse collection of reviews into an analytical dataset suitable for analyzing sentiment–rating incongruence across location, time, and reviewer behavior. The study used the "Tourism and Travel Reviews: Sri Lankan Destinations" dataset from Mendeley Data [17], which contains 16,156 reviews from 2010 to 2023 across 11 attraction types in Sri Lanka. Missing values and duplicates were checked during preprocessing. Date fields were used to create travel year and review delay, with negative delay values set to zero. Raw location text was processed using rule-based parsing and manual mapping to identify province and district. *Review_Length* was calculated as the character count of the review text. Star ratings were grouped into three classes:

- Negative (1–2⋆)
- Neutral (3⋆)
- Positive (4–5⋆)

This grouping follows common practice in sentiment analysis [1], [2] and makes the rating scale directly comparable with the three-class sentiment output. Table II presents the source columns used to build the analytical dataset and only the relevant columns used in analysis are presented.

TABLE II
SOURCE COLUMNS USED IN ANALYSIS

Feature	Description
Location	Raw location text used to derive province and district
Location_Type	Type of attraction (museum, beach, temple, etc.)
User_Contributions	Total number of reviews posted by the reviewer
Travel_Date	Date of the visit; used to derive travel year
Published_Date	Date of review posting; used to derive review delay
Rating	Star rating used to create Rating_Class
Title	Review title; combined with text for sentiment inference
Text	Review body; used for sentiment and review length

B. Sentiment Model Selection

To assign sentiment labels independently from star ratings, 4 transformer-based models were tested on a manually labeled set of 1,000 reviews. The dataset was split into 700 training and 300 testing instances, where the training portion was used to fine-tune selected models and the test set was used for comparative evaluation. Review titles and texts were combined as input, and model performance was assessed using Macro F1, accuracy, and weighted F1. As shown in Table III, the pretrained cardiffnlp/twitter-roberta-base-sentiment model performed best overall and was selected to label the full dataset [14], [15]. This model achieved a strong balance between classification performance and generalization, outperforming fine-tuned variants while avoiding potential overfitting given the limited size of the labeled dataset. This approach allowed sentiment to be identified from the review text rather than assumed from the rating.

TABLE III
SENTIMENT MODEL PERFORMANCE (TEST SET, $n = 300$)

Model	Acc.	Mac. F1	Wt. F1	Decision
gosorio/robertaSentimentFT	0.730	0.281	0.616	Rejected
cardiffnlp/twitter-roberta (pretrained)	0.830	0.692	0.805	**Selected**
Fine-tuned Cardiff RoBERTa	0.733	0.677	0.758	Lower overall fit
Fine-tuned RoBERTa-base	0.777	0.698	0.780	Higher Mac. F1 only

C. Variable Construction

After sentiment labeling, raw title and text were removed from further analysis. *Incongruent* was defined as a mismatch between *Sentiment* and *Rating_Class*, while *Pattern* recorded the six mismatch types. *Reviewer_Tier* grouped reviewers as follows: Novice (0–5), Casual (6–20), Active (21–100), Expert (101+).

Reviewer tiers were defined by analyzing the distribution of contributions. The data show strong positive skew (median $= 54$, max $= 9010$): most reviewers contribute 1–5 reviews, while few exceed 100. Based on this distribution, thresholds were set at 0–5, 6–20, 21–100, and 101+ to capture distinct engagement levels. Each tier corresponds to measurable differences in behavior. Reviewers with 0–5 reviews exhibit minimal platform familiarity, whereas the 6–20 range reflects casual engagement. The 21–100 tier identifies active users with sustained participation, and 101+ represents highly engaged expert reviewers. This tiering is further supported by rating behavior: the Conservative Rater pattern increases from 27.4% among novice reviewers to 40.4% among experts, indicating experience-dependent rating practices. These thresholds therefore reflect both the underlying data distribution and observable shifts in reviewer behavior.

The constructed analytical variables used in this study comprise both target-defining and explanatory features. *Sentiment* is represented as a three-class label derived from model output, while *Rating_Class* is a three-class label obtained by grouping star ratings (1–5); together, these variables define sentiment–rating incongruence, from which the binary variable *Incongruent* (0/1) is derived as the target outcome. *Pattern* is a six-category variable formed from the interaction between *Sentiment* and *Rating_Class*, capturing distinct mismatch typologies, and *Reviewer_Tier* is a four-level ordinal variable based on grouped contribution levels, used as a predictor. Continuous predictors include *log_review_length*, computed as the logarithm of review length in characters using $\log(1 + x)$, and *log_review_delay*, defined as the logarithm of the time gap between visit and posting using $\log(1+x)$. Temporal effects are modeled using *Travel_Year_c*, a centered travel year variable, and its squared term *Travel_Year_c*^{2}, which captures potential nonlinear time trends. Together, these variables provide a structured representation of textual, behavioral, and temporal factors relevant to the analysis.

D. Statistical Testing and Predictive Modeling

The final stage analyzed predictors of sentiment–rating incongruence using both statistical tests and predictive models. *Incongruent* was used as the binary outcome variable, while variables used to define it were excluded from the predictors to avoid data leakage. The final predictor set included venue type, province, reviewer tier, review length, review delay, and travel year terms, with low multicollinearity (max VIF $= 3.663$).

Chi-square and Mann–Whitney U tests were used for bivariate analysis, with Benjamini–Hochberg correction applied for multiple testing. Logistic regression and logit models were used to examine linear effects and adjusted odds ratios, while Random Forest was used to assess nonlinear relationships. SHAP was included as a supportive post hoc interpretation tool to summarize feature influence rather than as a primary modeling component. Model performance was evaluated using AUC-ROC.

TABLE IV
MODELING AND INTERPRETATION FRAMEWORK

Model	Purpose	Metric	Key Detail
1A: Logistic Reg.	Linear baseline performance	AUC-ROC	Stratified 80/20 split; balanced classes; 5-fold CV
1B: Logit (statsmodels)	Identifies independent drivers	95% CI	Unweighted inference model fit on full design matrix
2: Random Forest	Captures complex relationships	AUC-ROC	GridSearchCV on training set; same held-out test set
SHAP (post hoc)	Explains model predictions	Mean $\lvert\phi\rvert$	TreeExplainer on the best Random Forest

IV. RESULTS

A. Prevalence and Six-Pattern Typology

Each incongruence type reflects a directional mismatch between rating and sentiment polarity. Pattern names were derived from their defining characteristics: *Conservative Rater* pairs positive sentiment with neutral ratings; *Obligatory 5-Star* pairs neutral sentiment with positive ratings; *Frustrated Neutral* pairs negative sentiment with neutral ratings; *Polite Inflator* pairs negative sentiment with positive ratings; *Harsh Deflator* pairs neutral sentiment with negative ratings; and *Punitive Rater* pairs positive sentiment with negative ratings.

Among the reviews analyzed, 3,005 were identified as incongruent, giving an overall prevalence of 18.6%, or roughly one in five reviews. Fig. 1 further shows that incongruence follows a six-pattern typology. The two most common patterns, Conservative Rater (38.4%) and Obligatory 5-Star (28.3%), together account for 66.7% of all incongruent reviews. This indicates that rating–text mismatch is directionally structured rather than random. In addition, Frustrated Neutral and Polite Inflator account for a further 24.2% of incongruent cases, showing that negative sentiment is often paired with non-negative ratings. These findings suggest that incongruence reflects consistent behavioral patterns, supporting further analysis across reviewer and contextual factors.

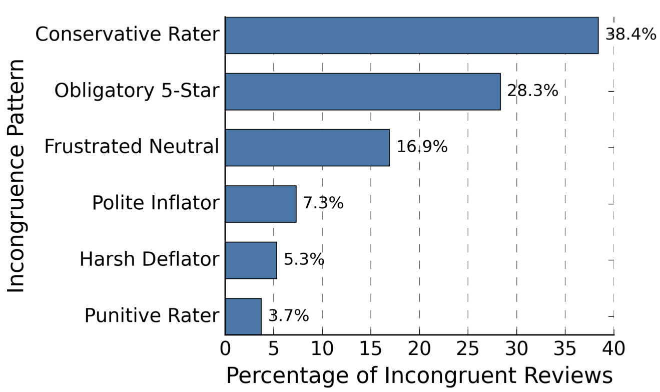

Fig. 1. Distribution of the six directional incongruence patterns.

B. Variation Across Venue Types

Incongruence rates were compared across 11 attraction categories to assess contextual variation. The Chi-square test showed a statistically significant association between venue type and incongruence, with a small but meaningful effect size ($\chi^2(10) = 125.85$, $p < 0.001$; Cramer's V = 0.088). As shown in Fig. 2, National Parks had the lowest incongruence rate (12.8%), whereas Museums had the highest (26.3%). Overall, the variation across venue types indicates that rating–text mismatch is not evenly distributed, but differs by review context.

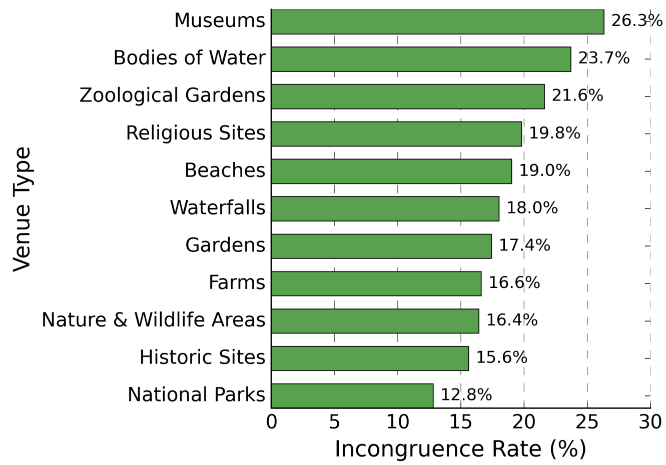

Fig. 2. Incongruence rate by venue type.

C. Predictors of Incongruence

Bivariate screening with Benjamini–Hochberg correction was used to identify predictors associated with incongruence. As shown in Table V, reviewer tier, province, travel year, and review length remained significant after correction, while review delay was not significant ($q = 0.7503$). Expert reviewers were 1.97 times more likely than novices to produce incongruent reviews. In addition, incongruent reviews were longer at the median than congruent reviews (296 vs. 279 characters). These findings indicate that both reviewer characteristics and review content are associated with sentiment–rating mismatch, although multivariable modeling is needed to test their independent effects.

Fig. 3 further illustrates the reviewer expertise effect for selected mismatch patterns, showing that Conservative Rater becomes more common among expert reviewers, while Harsh Deflator becomes less common.

TABLE V
BIVARIATE PREDICTOR SCREENING WITH BENJAMINI–HOCHBERG FDR CORRECTION

Predictor	Test Statistic	Raw p	q-value	Retained
Reviewer Tier	$\chi^2(3) = 85.99$	1.59×10^{-18}	7.94×10^{-18}	Yes
Province	$\chi^2(7) = 49.10$	2.17×10^{-8}	3.62×10^{-8}	Yes
Travel Year	Mann–Whitney	7.73×10^{-11}	1.93×10^{-10}	Yes
Review Length	Mann–Whitney	0.0011	0.0014	Yes
Review Delay	Mann–Whitney	0.7503	0.7503	No

D. Model-Based Analysis and Interpretation

1) Model 1A – Logistic Regression: A class-balanced logistic regression model was used as a linear baseline. It achieved a mean cross-validated AUC of 0.5890 ± 0.0093 and a test AUC of 0.5840, indicating modest but stable predictive performance. This suggests that incongruence is only partly explained by the observed variables.

2) Model 1B – Explanatory Logit: The explanatory logit model was used to identify independent predictors of incongruence. Based on 95% confidence intervals, 19 predictors were statistically significant. Venue type, reviewer expertise, review length, and travel year showed important effects, while review delay had only a weak negative association. Overall, the model shows that incongruence is shaped by structural, behavioral, and temporal factors.

3) Model 2 – Random Forest (Nonlinear Structure Test): A Random Forest model was applied to capture nonlinear relationships and interactions. It achieved a test AUC of 0.6095, which was slightly higher than logistic regression. This indicates that nonlinear effects are present, although their contribution is modest.

4) SHAP Analysis SHAP analysis was employed as a post hoc interpretability tool to examine the contributions of individual features in the Random Forest model. As tree based models capture nonlinear relationships and feature interactions, their internal decision structure is not directly interpretable. SHAP provides a principled, model agnostic framework for attributing predictions to input features, enabling consistent and locally accurate explanations. Using SHAP, key predictors such as review length, reviewer expertise, travel year, review delay, and certain venue types (e.g., museums) were identified as influential. The results were broadly consistent with those of the logit model, while also indicating that some effects vary across contexts and interactions. Overall, SHAP provides complementary insights into nonlinear behavior, supporting rather than replacing the primary inferences derived from the statistical and predictive models.

Temporal effects indicate a modest but consistent decline in sentiment–rating incongruence over time. The linear travel year term shows a significant negative association (Travel_Year_c: OR = 0.946, CI < 1), suggesting that more recent reviews are less likely to be incongruent. In contrast, the quadratic term (Travel_Year_c²: OR = 0.995) is not statistically significant, providing no evidence of nonlinear temporal effects and indicating an approximately linear trend. SHAP analysis further supports this finding, identifying Travel_Year_c as an influential predictor (mean |SHAP| = 0.0219) and confirming the overall direction of the effect.This suggests that, over time, reviewers are expressing their opinions more consistently, with written sentiment aligning more closely with their ratings.

V. Discussion

A. Structured Incongruence

An incongruence rate of 18.6% shows that rating-text mismatch is systematic. Ratings capture overall judgments; text captures specific details. Evaluation patterns differ by context. Religious sites illustrate this: high ratings reflect cultural or spiritual weight, while text documents specific issues accessibility, pricing, operations. Museums show similar patterns through different factors. They involve layered experiences that resist a single score. A guest rates 3 stars while describing strong exhibits but flagging way finding problems and crowding. The implication is practical. Star ratings don't function equally across contexts as sentiment proxies. Using them without validation introduces systematic bias into your models. Ratings and textual sentiment capture different dimensions of experience. Treating them as interchangeable signals creates modeling risk.

B. Conservative Rating Behavior

The Conservative Rater pattern further reinforces structural incongruence. Positive textual sentiment paired with moderate ratings accounts for 38.4% of incongruent reviews. Reviewers temper their numerical scores relative to their written sentiment rather than align them fully. Prevalence of this pattern increases from 27.4% (novice reviewers) to 40.4% (expert reviewers), showing that rating behavior becomes more calibrated with experience. Experienced reviewers apply stricter standards, reserving high ratings for exceptional experiences while maintaining positive text sentiment. This indicates incongruence reflects deliberate practice, not inconsistency. In short, sentiment based models misclassify positive experiences expressed through moderate ratings. Accounting for reviewer expertise is critical when interpreting rating-text relationships.

C. Location Type as a Structural Moderator of Label Reliability

Location type plays a clear role in how well ratings reflect sentiment. Some attractions show much higher levels of mismatch than others. For example, compared with national parks, museum reviews are over twice as likely to be incongruent (Adjusted OR = 2.386), with similar trends observed for beaches, inland waterbodies, zoological gardens, and waterfalls.

A simple interpretation is that certain attraction types are harder to evaluate with a single score. Experiences at museums or cultural sites are often more complex and multi-dimensional, which may lead reviewers to express mixed sentiment in text while assigning moderate or high ratings. In contrast, more straightforward experiences may produce closer alignment between ratings and sentiment.

These findings suggest that the reliability of star ratings is context-dependent, rather than uniform across all review types. As a result, treating all ratings as equally reliable sentiment labels may introduce systematic bias, highlighting the need for more context aware sentiment modeling.

D. Complementary Roles of Linear and Nonlinear Models

The results show that linear and nonlinear models offer complementary insights into sentiment–rating incongruence rather than competing explanations. Logistic regression (AUC = 0.584) provides a stable and interpretable baseline, identifying 19 significant predictors, while the Random Forest achieves a modest improvement (AUC = 0.6095), confirming the presence of nonlinear and interaction effects. However, the limited performance gain suggests that these nonlinearities are not dominant. The overall predictive range (AUC ≈ 0.58–0.61) indicates that incongruence is structured but only partially observable, with substantial variation driven by latent behavioral and contextual factors. Importantly, this moderate predictive performance should not be interpreted as a weakness; rather, it reflects the inherent complexity and subjectivity of human judgment in review behavior, where not all influencing factors are directly measurable. SHAP analysis reinforces this interpretation by showing that variables such as reviewer expertise, review length, and venue type contribute in nonlinear and context-dependent ways, underscoring the need for richer features to further capture the phenomenon.

E. Review Delay and Reviewer Expertise Effects

Review delay plays a limited role in sentiment–rating incongruence, showing no significance in bivariate analysis and only a weak negative association (OR = 0.960). In contrast, reviewer expertise is more influential, with experts exhibiting more conservative and fewer harsh rating behaviors, indicating that incongruence is driven more by reviewer behavior than timing.

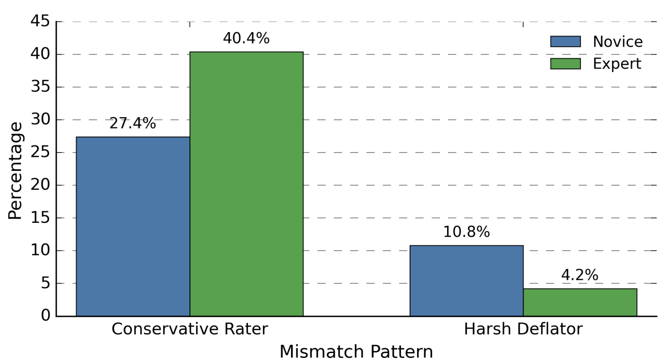

Fig. 3. Selected incongruence patterns by reviewer expertise, showing higher Conservative Rater prevalence and lower Harsh Deflator prevalence among expert reviewers.

VI. Conclusion

Sentiment-rating incongruence in tourism reviews is not random. The patterns we found are consistent and explainable. Reviewers approach rating and reviewing differently. Context shapes how they evaluate. Using transformer-based sentiment analysis with statistical and interpretable machine learning, we identified what drives the mismatch. The study reveals three key findings. First, we mapped different types of incongruence and measured how common they are. Second, we traced them back to reviewer expertise, review length and depth, and location type. Third, ratings and textual sentiment capture different aspects of the guest experience. They are not interchangeable signals. For practice, the takeaway is straightforward. Treating star ratings as ground truth without checking introduces bias. Rating-text misalignment is not noise to ignore. It is a signal. It shows how guests actually experienced and articulated their stay. We analyzed a large dataset from Sri Lanka, a tourism market that research overlooks. The framework likely transfers to other review platforms using similar structure. The approach scales well with structured features and machine-derived sentiment labels, though deeper semantic analysis could reveal more. The work opens two directions: test whether these patterns hold in other domains and contexts, and integrate richer linguistic and contextual features to better explain sentiment-rating divergence.

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